Step 2: From Complexity Metrics to the Dynamic Structure of the Subject

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Summary

Building on the Σ* metric that identifies regimes of organized neural complexity, this framework defines a methodological program to determine whether such regimes reflect stable dynamic attractors in neural state space. It integrates window selection, geometric reconstruction, stability analysis, recurrence, and robustness, as well as falsifiable predictions about consciousness dynamics. The conceptual transfer stage is reserved for Step 3, conditional on empirical validation of attractors.

1. Objective

The goal of this step is to extract operational information about the existence, properties, and behavior of potential neural attractors, so that findings can be integrated into a coherent dynamic model of the subject as an attractor. Specific objectives include:

1. Confirm that high Σ* regions reflect organized activity.

2. Determine the geometry and effective dimensionality of reconstructed attractors.

3. Evaluate robustness against perturbations and trajectory coherence.

4. Generate falsifiable predictions on neural dynamics associated with consciousness.

2. Conceptual Pipeline

2.1 Dynamic Window Selection

Input: Temporal segments with high Σ^*.

Action: Verify internal consistency of Σ^*, discarding isolated peaks.

Output: Representative windows of organized activity, ready for geometric reconstruction.

2.2 Reconstruction and Visualization

Methods: Multichannel embedding (extended Takens), dimensionality reduction (PCA, diffusion maps, autoencoders).

Output: Reconstructed space with compact, recurrent manifolds.

[Conceptual diagram: reconstructed manifold illustrating recurrent attractor regions]

2.3 Geometry and Dimensionality Analysis

Metrics: Fractal dimension, Kaplan–Yorke, PCA participation ratios, local curvature, persistent homology.

Interpretation: Compact and recurrent subspaces indicate candidate attractors.

[Conceptual illustration: fractal dimension and PCA participation ratios]

2.4 Stability and Convergence Evaluation

Trajectories from diverse initial conditions → dynamic clustering.

Lyapunov exponents near zero or negative → stability or meta-stability.

Interpretation: Robust convergence and functional stability of attractors.

[Conceptual schematic: Lyapunov trajectories showing stability patterns]

2.5 Recurrence and Temporal Topology

Recurrence plots and persistent homology.

Interpretation: Recurrent sequences reinforce evidence for coherent dynamic structures.

[Conceptual diagram: recurrence plot illustrating temporal structure]

2.6 Robustness to Perturbations

Potential experiments: TMS, sensory/electrical stimulation.

Measurement: Temporal deviations and return time to attractor basin.

Interpretation: Recovery capacity indicates attractor depth and stability.

[Conceptual schematic: recovery trajectory demonstrating attractor robustness]

2.7 Relation to Cognitive States and Transitions

Compare attractors across wake/sleep, attention/distraction, pre-anesthesia/anesthesia. Changes in geometry, dimensionality, or stability reflect transitions between attractors.

[Conceptual illustration: geometry and dimensionality changes across cognitive states]

2.8 Integration with Σ*

High Σ* windows → recurrent subspaces + trajectory convergence + resilience.
Outcome: mapping Σ* ↔ dynamic attractor.

[Conceptual diagram: mapping Σ* onto attractor stability metrics]

3. Falsifiable Predictions

1. Convergent trajectories in high Σ* windows → existence of attractors.

2. Perturbations → systematic temporal deviations with recovery.

3. Cognitive transitions → geometric changes in reconstructed space.

4. Multichannel coordination → consistency in effective dimensionality and recurrence.

5. Robustness → Lyapunov exponents near zero or negative.

4. Conceptual Formalization

Let be the attractor associated with each Σ*_i window:

Dynamic confinement: compact and recurrent.

Trajectory convergence: convergence for initial conditions .

Robustness to perturbations: temporal deviations corrected within Δt.

Relation with Σ:* High Σ* ↔ strong attractor properties.

5. Outputs and Applications

1. Neural attractor maps: regions in reconstructed space with defined dynamic properties.

2. Stability indicators: dimension, Lyapunov, recurrence, persistent topology.

3. Predictions of state transitions: models of changes between attractors.

4. Conceptual basis for hypothetical transfer: Step 3, conditional on attractor validation.

6. Conceptual Figure (Pipeline Summary)

NEURAL ACTIVITY (EEG/MEG)

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Σ^* Calculation → Regime Detection

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Temporal Segmentation (high Σ^* windows)

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Space Reconstruction (Multichannel Takens Embedding)

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Reconstructed Space $Y(t) \in \mathbb{R}^D$

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Geometric & Dynamic Analysis:

• Fractal Dimension / Kaplan–Yorke

• Lyapunov Exponents

• Recurrence

• Persistent Topology

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Identification of Dynamic Structure:

Compact manifolds, recurrent regions

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Attractor Evaluation:

• Trajectory convergence

• Stability under perturbations

• Attractor basin existence

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Conclusion: Evidence of a reflexive dynamic attractor

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(Step 3: Conceptual Transfer, only if attractor validated)